

# Multi-Layer Dos Detection and Severity Scoring System Using Hybrid ML and DL Approaches.

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Project Proposal Report

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## DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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# Abstract

Denial of Service (DoS) attacks are still one of the most serious threats to network availability. Among these attacks, UDP Flood attacks make up a large share of volumetric network layer attacks. In a UDP Flood attack, the attacker takes advantage of the connectionless nature of the User Datagram Protocol (UDP). The attacker sends a very large number of packets to a target system. Because UDP does not require a connection setup, the system must process every packet it receives, even if the packet is malicious.

This behavior quickly consumes important system resources. The CPU processing capacity, memory buffers, and network bandwidth begin to overload. As the traffic continues, the service becomes slow and may eventually stop responding to legitimate users. Modern Intrusion Detection Systems (IDS) can detect these attacks with reasonable accuracy. However, most existing systems only provide a binary output, which means the traffic is classified as either normal or attack. They do not show how serious the attack is. Because of this limitation, Security Operations Centre (SOC) analysts may struggle to prioritise incidents when many alerts appear at the same time.

This research introduces a severity aware UDP Flood detection component. It is developed as part of the Multi Layer DoS Detection and Severity Scoring System (Project R26 CS 016). The overall project studies four major DoS attack types. These are ICMP Flood, UDP Flood, SYN Flood, and Fragmentation attacks.

The proposed system uses a hybrid detection architecture that combines Machine Learning (ML) and Deep Learning (DL) techniques. Three ML classifiers are evaluated in this study. These are Random Forest (RF), Support Vector Machine (SVM), and AdaBoost. Their performance is compared with a hybrid deep learning model that combines Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks. The models are trained and tested using network flow features taken from well-known benchmark datasets such as CICIDS2017 and CICDDoS2019.

The main contribution of this research is the introduction of a severity scoring mechanism that measures the operational impact of detected UDP Flood attacks. The design of this scoring system is inspired by the Common Vulnerability Scoring System (CVSS v3.1), which is widely used to measure vulnerability severity. In this system, the severity score is calculated using network traffic intensity.

The severity score is calculated using the following formula:

$$\text{Severity Score} = \text{Flow Packet Count} \times 2.5$$

The Flow Packet Count represents the total number of packets in a network flow. The weight value 2.5 reflects the strong impact of UDP Flood attacks on network availability when compared with other DoS attack vectors in the project.

After the score is calculated, it is mapped into four operational severity levels. These levels are Low, Medium, High, and Critical. This classification helps security analysts understand how urgent an attack is and decide the appropriate response.

The calculated severity results are then sent to the Final Severity Aggregator. This module combines the outputs from all four project components and produces a unified DoS attack severity ranking framework.

By combining hybrid ML and DL detection with a CVSS inspired severity scoring method, this research addresses an important limitation in current DoS detection systems. The approach helps security teams better prioritise threats and improve incident response in modern network security environments.

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## LIST OF ABBREVIATIONS

| Abbreviation | Full Form                                  |
|--------------|--------------------------------------------|
| AI           | Artificial Intelligence                    |
| AHP          | Analytic Hierarchy Process                 |
| API          | Application Programming Interface          |
| AVSF         | Attack Vector Severity Factor              |
| CIC          | Canadian Institute for Cybersecurity       |
| CNN          | Convolutional Neural Network               |
| CVSS         | Common Vulnerability Scoring System        |
| DL           | Deep Learning                              |
| DoS          | Denial of Service                          |
| DDoS         | Distributed Denial of Service              |
| DSR          | Design Science Research                    |
| IDS          | Intrusion Detection System                 |
| IAS          | Information Assurance and Security         |
| LSTM         | Long Short-Term Memory                     |
| MCDA         | Multi-Criteria Decision Analysis           |
| ML           | Machine Learning                           |
| NIDS         | Network Intrusion Detection System         |
| RFC          | Request for Comments                       |
| REST         | Representational State Transfer            |
| RF           | Random Forest                              |
| SHAP         | SHapley Additive Explanations              |
| SIEM         | Security Information and Event Management  |
| SMOTE        | Synthetic Minority Over-sampling Technique |
| SOC          | Security Operations Center                 |
| SVM          | Support Vector Machine                     |
| UDP          | User Datagram Protocol                     |
| WSM          | Weighted Sum Model                         |



# 1. INTRODUCTION

## 1.1. Background and Context

Denial of Service (DoS) attacks are threats at the network level that make a service unavailable by flooding a target system with bad traffic, making it impossible for real users to access it. A DoS attack can come from one or a few sources, while a Distributed Denial of Service (DDoS) attack comes from many coordinated botnet nodes. This makes DoS attacks just as dangerous but much easier to start with little infrastructure. This project deals with both single-source DoS and distributed volumetric attacks because the detection models work on traffic behavior patterns that work for both. This study constitutes Component 2 of the Multi-Layer DoS Detection and Severity Scoring System (Project R26-CS-016), which tackles four attack vectors - ICMP Flood, UDP Flood, SYN Flood, and Fragmentation attacks - within a cohesive severity ranking framework. This part does UDP Flood Detection and Severity Analysis, and the weighted severity output goes into the group's Combined Scoring Ranking Table.

### 1.1.1 The Real-World Problem: UDP Flood Attacks on Network Infrastructure

Modern organizations in banking, healthcare, government, and telecommunications rely on stable network connectivity. Many of their daily operations depend on online systems. If the network becomes unavailable, important services may stop working. This can affect customers, staff, and critical infrastructure.

In this project, four DoS attack vectors are studied. These are ICMP Flood, UDP Flood, SYN Flood, and Fragmentation attacks. Among them, UDP Flood attacks are especially harmful because of the basic design of the User Datagram Protocol (UDP).

In a UDP Flood attack, an attacker sends a very large number of UDP packets to different ports on a target host. UDP is connectionless and stateless. It does not perform a handshake and it does not keep session information. Because of this, the target system must process every packet it receives, even if the packet is not legitimate.

If no application is listening on the targeted port, the operating system may respond with an ICMP “Destination Unreachable” message. This depends on the system configuration and rate limiting settings. Creating these responses consumes CPU resources and outgoing bandwidth.

If a service is running on the port, such as DNS on UDP port 53, the situation becomes different. In this case, the flood directly overloads the application’s processing queue. No ICMP response is generated, but the service becomes overwhelmed by the high number of incoming packets.

When the packet rate remains high for a long time, the system resources start to fail. The CPU becomes overloaded, memory buffers fill up, and network bandwidth becomes saturated. At this point the host may stop responding, making the service unavailable to legitimate users.

The barrier to launching this attack is very low. Tools such as hping3 and UDP Unicorn allow attackers to generate large traffic floods with very little setup. There are also stress testing services that can be misused to launch attacks. Some of these services allow attacks to be launched for as little as USD 10 per hour.

Because of this easy access, UDP Flood attacks are among the most common volumetric DoS attacks seen today. This growing threat highlights the need for hybrid ML and DL based detection systems that can also measure the severity of an attack. Such systems can help security teams detect attacks faster and respond based on how serious the traffic flood is.

### 1.1.2 Affected Stakeholders and the Significance of the Problem

UDP Flood attacks affect many groups in different ways. In Sri Lanka, the telecommunications providers are often the most affected. Companies such as Dialog Axiata, SLT, and Mobitel depend on stable network traffic to deliver services. When a UDP flood occurs, large volumes of traffic can congest upstream transit links. This causes network slowdowns and service interruptions for thousands of users at the same time.

Recent data shows how serious this problem has become. In 2024, the telecom sector accounted for about 14% of all DoS and DDoS attacks in the Asia Pacific region. This was an 86% increase compared to the previous year.

Studies also show that UDP Flood and DNS Amplification attacks made up about 45% of telecom related attacks in the region [18]. These attacks create major operational problems for service providers.

The financial impact is also significant. For financial institutions, the average cost of a single incident can reach around USD 1.35 million [13]. Service outages may interrupt online banking systems and payment platforms. This affects both businesses and customers who depend on these services.

Government services are also vulnerable. When online portals become unavailable due to DoS attacks, people may lose confidence in digital government systems. This weakens public trust in online services that are meant to improve efficiency and accessibility.

The healthcare sector faces even more serious risks. Hospitals rely on network connectivity to share patient records and communicate between systems. If DoS attacks interrupt these connections, medical staff may lose access to important information. In critical situations, this can delay treatment and put lives at risk.

Cloud service providers also experience challenges because their infrastructure supports many tenants at the same time. A large attack can affect several services that share the same network resources. At the same time, Security Operations Centre (SOC) analysts face another problem called alert fatigue. Many current IDS tools generate repeated alerts without showing how serious each attack is. Studies show that about 67% of analysts struggle to handle the high number of alerts produced by these systems.

This is why severity scoring becomes important. It helps security teams understand which attacks require immediate attention and which ones are less critical. In countries like Sri Lanka, where about 52% of the population used the internet in 2023 [27], protecting network services is increasingly important for daily life, business operations, and public services.

### 1.1.3 Statistical Evidence: Global, Regional, and Local

| Scope            | Key Statistic                                                                                                                                                                                       | Source                         | Year    |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|---------|
| Global DoS/DDoS  | Volumetric DoS/DDoS attack frequency increased 117% year-on-year; network-layer attacks constitute most of all incidents                                                                            | Cloudflare [14]                | 2024    |
| Global DoS/DDoS  | Registered attacks grew from ~274,000 (Q1 2023) to ~512,000 (Q4 2024) — an 87% rise in two years; majority are volumetric DoS-style floods                                                          | Statista [22]                  | 2024    |
| Global DoS       | Average enterprise cost of a volumetric DoS attack: USD 50,000 per hour; financial sector average: USD 1.35 million per incident                                                                    | NETSCOUT [13]                  | 2023    |
| Global UDP Flood | UDP Flood is the #1 network-layer (L3/L4) DoS attack vector globally by packet volume, constituting 35–40% of all volumetric attack traffic                                                         | NETSCOUT [13]; Cloudflare [14] | 2023–24 |
| APAC             | Asia-Pacific DoS/DDoS attacks surged 92% in 2024; a further 116% YoY in H1 2025, with botnet sizes quadrupling — majority executed as volumetric UDPstyle floods                                    | Storm Wall [18]                | 2025    |
| APAC Telecom     | Telecom sector targeted in 14% of all APAC attacks (+86% YoY); UDP Flood + DNS Amplification = 45% of all telecom-targeted incidents                                                                | Storm Wall [18]                | 2025    |
| South Asia       | India targeted by 25+ hacktivist groups using UDP Flood techniques (OpIndia campaign); Bangladesh recorded 71% DoS volume spike during Q1 2024 elections                                            | Storm Wall [18]; CERT-In [24]  | 2024–25 |
| Sri Lanka        | SLCERT CC recorded 596 cyber incidents in 2019, rising to 4,347 in 2024 — a 629% increase; formal reporting estimated at ~30% of actual incidents, indicating severe under-reporting of DoS attacks | Lenasia / SLCERT [17]          | 2026    |

|           |                                                                                                                                                                                                                         |                          |      |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|------|
| Sri Lanka | Cybersecurity market projected at USD 40.63 million by 2025; National Cyber Security Strategy 2024– 2027 explicitly identifies DoS/DDoS as a priority infrastructure threat requiring detection capability improvements | Statista [22]; MoDE [21] | 2025 |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|------|

Table 1Table 1 Statistical Evidence: Global, Regional and Local DoS Attack Statistics

### 1.1.4 Alignment with the United Nations Sustainable Development Goals (SDGs)

| SDG                | Goal Title                              | Relevant Targets & Indicator                                                                                                                                                            | Research Contribution                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------------|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SDG 9<br>(Primary) | Industry, Innovation and Infrastructure | Target 9.1: Resilient, sustainable infrastructure.<br>Target 9.c: Universal ICT access.<br>Indicator 9.c.1: Mobile network coverage by technology.                                      | UDP Flood DoS attacks directly degrade ICT infrastructure reliability — the core concern of Target 9.1. This component produces an opensource, deployable detection system that ISPs and network operators can adopt without commercial licensing costs. Sri Lanka's National Cyber Security Strategy 2024–2027 explicitly identifies DoS/DDoS resilience as a prerequisite for its SDG 9 commitments. |
| SDG 16             | Peace, Justice and Strong Institutions  | Target 16.6: Accountable, transparent institutions.<br>Target 16.10: Public access to information.<br>Indicator 16.10.2: Countries with legal guarantees for public information access. | Government portals and e-governance platforms are primary targets for DoS attacks in South Asia. UDP Flood-induced unavailability denies citizens access to judicial records and official information, undermining Target 16.10. Severity-aware detection enables institutions to respond proportionately and maintain availability.                                                                   |
| SDG 17             | Partnerships for the Goals              | Target 17.6: Global knowledge-sharing in science and innovation.<br>Indicator 17.6.2: Fixed broadband subscriptions per 100 inhabitants.                                                | Open-source release of the hybrid ML/DL detection toolkit and an IEEE-indexed publication contribute to the global knowledge commons. Protecting broadband infrastructure from UDP Flood DoS attacks directly advances Indicator 17.6.2 in emerging economies including Sri Lanka.                                                                                                                     |

Table 2Table 2 SDG Alignment and Research Contribution

This research mainly supports **Sustainable Development Goal 9**, which focuses on building strong and reliable infrastructure. Today, digital infrastructure is an essential part of modern society. Banking systems, healthcare services, government platforms, and communication networks all depend on stable internet connectivity. Protecting these systems from DoS attacks helps ensure that these services remain available and reliable.

The study also relates to Sustainable Development Goal 16, which focuses on building strong institutions and improving public access to information. When DoS attacks disrupt government websites or online public services, citizens may lose access to important information. This limits transparency and reduces trust in digital government systems.

In addition, this research supports Sustainable Development Goal 17, which encourages global partnerships and knowledge sharing. By publishing the findings through open research, the study can help researchers, engineers, and cybersecurity professionals around the world better understand DoS attack detection and severity analysis. This helps improve global knowledge and supports stronger digital security practices.

### 1.1.5 Hybrid ML and DL as the Technical Response

Traditional Intrusion Detection Systems (IDS), including Snort and Suricata, predominantly utilize signature matching and rule-based filtering techniques. These systems scrutinize network traffic, comparing it to known attack patterns stored in predefined rule sets. Although this method is successful in identifying established threats, it encounters limitations when dealing with UDP Flood Denial of Service (DoS) attacks.

First, systems that use signatures have difficulty detecting new or zero-day attack patterns. If the attack pattern isn't already in the rule database, the system might not be able to find it. Second, these systems require constant manual updates to their rules.

Security teams must regularly create and maintain rules, which can be time consuming and difficult at scale. Third, when a rule is matched, the system usually generates the same binary alert. It simply reports normal or attack, without explaining how serious the attack is.

These limitations become more visible in this multi component research project, where four different DoS attack vectors are studied. Each attack type may require different traffic features and detection methods.

To address these issues, this study proposes a hybrid detection architecture that combines Machine Learning (ML) and Deep Learning (DL). This hybrid design is also the key innovation described in the overall project title.

The ML stage uses three classifiers: Random Forest (RF), Support Vector Machine (SVM), and AdaBoost. These models are easier to interpret and have relatively low processing latency. They can detect patterns in UDP Flood traffic using labeled CICFlowMeter flow features.

The DL stage uses a CNN and LSTM hybrid model. This model can capture both spatial and temporal patterns in high volume network traffic. Such patterns are often difficult for traditional ML classifiers to model.

Both pipelines are evaluated side by side during experiments. The model that shows the best performance is used to support the severity scoring engine. In addition, ML classifiers provide feature importance values, which help refine the severity scoring formula. At the same time, the DL model shows strong performance in detecting temporal attack patterns in the CICDDoS2019 dataset.

### 1.1.6 Research Cluster Alignment

This component connects with five academic research clusters. Three of them are the main focus areas. Two others support the work and strengthen the research direction.

- The first main cluster is Network Intrusion Detection Systems (NIDS). In this study, the system uses flow level anomaly detection to identify UDP Flood DoS traffic. The models learn traffic patterns using benchmark datasets such as CICIDS2017 and CICDDoS2019. These datasets provide structured network flow features that help train and evaluate the detection models.
- The second main cluster is Hybrid ML and DL for Cybersecurity. This part compares three Machine Learning classifiers. These are Random Forest, Support Vector Machine, and AdaBoost. Their performance is tested against a CNN LSTM hybrid deep learning model. The system also uses SHAP based explainability. This helps explain how the models make decisions and which features influence the results.
- The third main cluster is Threat Intelligence and Risk Quantification. This research introduces a weighted severity scoring engine that follows the TAF framework used in the project. The severity score is calculated using a simple formula.

$$\text{Severity Score} = \text{Flow Packet Count} \times 2.5$$

This scoring method provides a continuous risk signal, similar to how CVSS scores represent vulnerability severity.

- There are also two supporting research alignments. The first is DoS Attack Analysis and Mitigation. This includes detailed UDP Flood feature analysis and sending the calculated severity output to the system's Final Aggregator.
- The second supporting area is Sustainable Digital Infrastructure. The research supports SDG 9, SDG 16, and SDG 17. It also promotes open source deployment for South Asia, so researchers and organizations in the region can benefit from the results.

| Research Cluster                               | Alignment | Primary Contribution                                                                                         |
|------------------------------------------------|-----------|--------------------------------------------------------------------------------------------------------------|
| Network Intrusion Detection Systems (NIDS)     | Primary   | Flow-level UDP Flood DoS detection on CICIDS2017/CICDDoS2019 using CICFlowMeter features                     |
| Hybrid ML and DL for Cybersecurity (ML/DL-Sec) | Primary   | RF, SVM, AdaBoost + CNN-LSTM hybrid evaluation with SHAP explainability and TAF-aligned severity scoring     |
| Threat Intelligence & Risk Quantification      | Primary   | TAF-aligned severity score (Flow Packet Count $\times$ 2.5) with four-tier output (Low/Medium/High/Critical) |

|                                              |            |                                                                                         |
|----------------------------------------------|------------|-----------------------------------------------------------------------------------------|
| DoS Attack Analysis & Mitigation             | Supporting | UDP Flood feature-level analysis; severity output feeds group Final Severity Aggregator |
| Sustainable Digital Infrastructure / SDG ICT | Supporting | Open-source deployable system for Sri Lanka / South Asia; SDG 9, 16, 17 aligned         |

Table 3 Table 3 Research Cluster Alignment

This part fits in with the IAS—Information Assurance and Security research group at SLIIT. It works well with the ICMP (C1), SYN (C3), and Fragmentation (C4) parts of the group project to make a single, multi-vector DoS detection and severity scoring system.

## 1.2 Literature Review

This section looks at current detection tools that are already in use and important research on using ML and DL methods for DoS detection. It looks at the methods, datasets, performance results, and limitations that led to the hybrid design of this part.

### 1.2.2 ML-Based DoS and UDP Flood Detection

Sharafaldin and colleagues (2018) introduced the CICIDS2017 dataset, which later became widely used in intrusion detection research. In their study, they tested several classifiers such as Random Forest (RF), Decision Tree (DT), and Naive Bayes. The results showed detection rates above 98 percent for DoS attack categories. Their work helped establish the CICFlowMeter feature schema as a common benchmark for Network Intrusion Detection Systems (NIDS) research. However, the system only produced binary outputs, meaning it could identify an attack but could not measure its severity [1].

Haider and his team (2020) proposed an ensemble intrusion detection system that combined AdaBoost and Random Forest algorithms. Their experiments used the NSL-KDD dataset. The system achieved an F1 score of 97.4 percent for detecting DoS attacks. The authors also addressed class imbalance by applying the SMOTE technique during training. Even though the detection performance was strong, the study did not include any method to measure the severity of attacks [2].

Jain and Bhandare (2021) studied UDP Flood traffic using the CICDDoS2019 dataset. They tested five machine learning classifiers, including Support Vector Machine (SVM) and K Nearest Neighbour (KNN). Their results showed that SVM achieved a precision of 96.8 percent in detecting UDP Flood attacks. However, the model required significant computational resources. The authors suggested that future systems should focus on models that are easier to deploy. They also noted that the lack of severity output remained a major limitation in existing approaches [3].

Agrawal and colleagues (2022) explored feature importance guided Random Forest models using the CICIDS2017 dataset. Their study showed that a reduced feature set of about 15 attributes could achieve performance similar to models that use the full feature set. Important features included packet rate, flow duration, and inter arrival statistics. This approach significantly reduced training cost while maintaining strong detection accuracy. Their findings provide useful guidance for feature selection in this component [5].

Mousavi and St Hilaire (2020) investigated early detection of DoS and UDP Flood attacks using entropy based techniques. Their method achieved detection latency below 500 milliseconds, but the system produced a false positive rate of about 12 percent. Based on these results, the authors suggested combining statistical detection methods with machine learning models to improve both speed and accuracy. This idea supports the use of hybrid detection architectures in later research [6].

### 1.2.3 DL Approaches for DoS Traffic Classification

Yin et al. (2019) utilized LSTM for DoS/DDoS detection in SDN, attaining significant temporal precision for persistent attack patterns while recognizing computational overhead as an impediment to edge deployment [4].

Rezaei and Liu's 2019 study assessed the efficacy of CNN architectures in classifying network traffic, confirming their capacity for strong spatial pattern recognition; however, they cautioned against overfitting, particularly when dealing with imbalanced datasets

[7]. This issue was directly addressed within this component through the implementation of SMOTE and stratified cross-validation techniques.

Ferrag and his team (2024) studied how well CNN-LSTM hybrid and transformer models performed on the CICDDoS2019 dataset. This dataset included various types of DoS/DDoS attacks, such as UDP Flood attacks. CNN-LSTM hybrids outperformed standalone architectures, achieving F1-scores above 97% for volumetric

DoS detection at lower inference latency than transformers [28]. The most recent deep learning study directly related to this component confirms the CNN-LSTM hybrid as the main deep learning architecture, and it also provides a quantitative benchmark for comparison.

### 1.2.4 Multi-Stage and Severity-Aware Systems

Tan et al. (2021) put forward a multi-stage XGBoost + rule-based framework for detecting DoS on CICDDoS2019, which had a 99.1% accuracy rate and a 0.3% false alarm rate. The authors clearly stated that their system's lack of a severity scoring component was a problem [8].

Kshirsagar and Kumar (2021) examined DoS detection datasets and metrics, contending that binary precision and recall are inadequate for operational IDS evaluation and promoting metrics for detection latency, resource utilization, and alert prioritization [9] all of which are covered in this component's evaluation plan.

The literature reviewed shows that binary classification accuracy is consistently high on CICIDS2017 and CICDDoS2019. Some SIEM platforms and commercial security tools use risk scores at a high level, but very few IDS systems offer quantitative, flow-level severity scoring for UDP Flood DoS attacks. None of these systems combine this with hybrid ML/DL detection, SHAP-based explainability, and SOC-compatible tiered alert output in a single deployable architecture. Section 1.3 formalizes the four gaps that led to this component.

## 1.3 Research Gap

A review of current DoS detection research and existing IDS tools shows several important gaps. These gaps explain why a new approach is needed in this study.

The proposed component introduces a novel approach by integrating three capabilities that are rarely combined in existing research. First, the system evaluates both machine learning and deep learning models in a hybrid detection framework for UDP Flood traffic. Second, the system introduces a quantitative severity scoring mechanism that converts detected attacks into operational risk tiers. Third, SHAP-based explainability is used to identify the most influential network features contributing to each detection decision.

While previous studies have focused primarily on improving classification accuracy, most existing research does not provide severity-aware outputs or explainable model behaviour. As a result, SOC analysts often receive large volumes of binary alerts without any prioritisation guidance.

By combining hybrid ML/DL detection, quantitative severity scoring, and explainable feature attribution within a single framework, the proposed system aims to provide more actionable alerts that support faster incident triage and response.

### Gap 1. Severity quantification is missing in most DoS detection systems:

Most intrusion detection research focuses on detection accuracy. The systems usually produce only a binary output such as normal or attack. For example, Sharafaldin et al. (2018) reported detection accuracy above 98 percent using the CICIDS2017 dataset, but the system did not measure attack severity [1]. Tan et al. (2021) also reported high detection accuracy of 99.1 percent but noted that severity scoring was not included in their system [8]. In real Security Operations Center (SOC) environments, binary alerts create practical problems. Analysts often receive a large number of alerts and cannot easily identify which incidents are more serious. Studies show that about 67 percent of SOC analysts struggle with alert fatigue because alerts do not provide severity information [25].

### Gap 2. Limited use of hybrid ML and DL architectures for UDP Flood detection:

UDP Flood attacks represent about 35 to 40 percent of global volumetric DoS traffic [13][14]. Despite this, many studies use either machine learning models or deep learning models, but rarely both together. Jain and Bhandare (2021) tested several ML classifiers on CICDDoS2019 UDP Flood data, but their work did not include deep learning models [3]. On the other hand, Ferrag et al. (2024) showed strong results using a CNN LSTM architecture, but they did not compare the model with traditional ML classifiers or include severity scoring [28]. At present, very few studies combine ML and DL models in one detection framework specifically for UDP Flood attacks.

### Gap 3. Limited explainability in DoS detection models:

Deep learning models often act like black boxes. They provide strong prediction results but give little explanation about why a decision was made. For instance, Ferrag et al. (2024) showed that CNN LSTM models perform well, but their system did not include any explainability method [28]. Without explanation, SOC analysts may find it

difficult to trust automated detection results. This is especially important in regulated environments where decisions must be justified. Techniques such as SHAP feature attribution are still rarely used in DoS detection research.

#### **Gap 4. Lack of severity based alerts for SOC and SIEM environments:**

Most detection systems generate alerts without prioritizing them by attack intensity. These binary alerts are difficult to integrate into Security Information and Event Management (SIEM) or Security Orchestration, Automation and Response (SOAR) workflows. This issue is even more serious in countries with limited cybersecurity resources. In Sri Lanka, for example, SLCERT|CC reported a 629 percent increase in cyber incidents between 2019 and 2024 [17]. In such environments, analysts need alerts that clearly show which attacks require immediate attention.

To address these gaps, this research proposes a hybrid detection framework that combines Machine Learning and Deep Learning models with a quantitative severity scoring mechanism. The system evaluates three ML classifiers Random Forest, Support Vector Machine, and AdaBoost together with a CNN LSTM deep learning model using the CICIDS2017 and CICDDoS2019 datasets.

In addition to detection, the system introduces a severity scoring formula:

$$\text{Score} = \text{Flow Packet Count} \times 2.5$$

This score allows each detected UDP Flood attack to be placed into four operational levels: Low, Medium, High, and Critical. The framework also uses SHAP based explainability to show which features influence each prediction. Finally, the system provides severity tiered alert outputs through a REST API, allowing integration with SOC platforms and the project's Final Severity Aggregator.

The novelty of this research lies in combining three capabilities within a single UDP Flood detection framework. First, the system evaluates both Machine Learning and Deep Learning models in a hybrid architecture to identify the most effective detection approach. Second, it introduces a quantitative severity scoring mechanism inspired by CVSS principles, allowing detected attacks to be classified into operational severity levels such as Low, Medium, High, and Critical. Third, the framework integrates explainable AI using SHAP to identify the key traffic features influencing detection decisions. By combining detection, severity assessment, and explainability, the proposed system provides more informative and actionable alerts for Security Operations Centre (SOC) analysts.

## **OBJECTIVES**

### **2.1 Main Objective**

The main objective of this study is To design, implement, and experimentally evaluate a severity-aware UDP Flood detection system using hybrid Machine Learning and Deep Learning models that can accurately detect attack traffic and classify attack severity levels using benchmark datasets such as CICIDS2017 and CICDDoS2019.. The system aims to accurately detect malicious network flows and assign a quantitative severity score. This score helps Security Operations Center (SOC) analysts prioritize incidents and respond more effectively.

### **2.2 Specific Objectives**

1. To study UDP Flood attack traffic behavior by extracting and analyzing network flow features from the CICIDS2017 and CICDDoS2019 datasets using CICFlowMeter.
2. To perform feature engineering and feature selection in order to identify the most relevant network flow attributes for UDP Flood detection. This process uses Random Forest feature importance, Chi square testing, and Pearson correlation analysis.
3. To implement and evaluate several Machine Learning and Deep Learning models, including Random Forest (RF), Support Vector Machine (SVM), AdaBoost, and a CNN LSTM hybrid architecture, for detecting UDP Flood attacks.
4. To compare the performance of these models using standard metrics such as Accuracy, Precision, Recall, F1 Score, ROC AUC, and False Positive Rate. This evaluation helps identify the best performing detection model.
5. To design a weighted UDP Flood severity scoring mechanism that assigns a numerical score to each detected attack using the formula:

$$\text{Score} = \text{Flow Packet Count} \times 2.5$$

This score allows each attack to be classified into four severity levels: Low, Medium, High, and Critical.

6. To apply explainable AI techniques using SHAP feature attribution and develop a prototype alert output system. This system will provide severity tiered alerts that can be integrated with SOC dashboards and the group project's Final Severity Aggregator.

## 2. METHODOLOGY

### 3.1 Research Approach

This research uses an Experimental and Design Science Research (DSR) methodology. This approach is suitable because the study focuses on designing and evaluating a new IT artefact. The artefact in this research is the hybrid ML and DL detection system with a severity scoring mechanism. The system is designed to address a clear operational problem in network security.

The Design Science approach guides the process of building and testing the proposed solution. It helps ensure that the system is not only theoretical but also practical and useful in real security environments.

The experimental part of the research focuses on quantitative evaluation. Four detection models are tested using well known benchmark datasets. The evaluation uses controlled experiments, statistical cross validation, and comparative analysis. The results are also compared with published baseline studies to measure the performance of the proposed system.

### 3.2 Dataset Collection

The first dataset is CICIDS2017, created by the Canadian Institute for Cybersecurity in 2017. It contains five days of labelled network traffic and includes 14 attack categories, including DoS attacks. The dataset has about 2.8 million flow records. In this research, UDP Flood traffic is taken from the DoS and DDoS category. The dataset also provides 83 precomputed CICFlowMeter features, which are commonly used in intrusion detection research [1].

The second dataset is CICDDoS2019, released by the Canadian Institute for Cybersecurity in 2019. This dataset includes 12 attack categories, with specific classes for UDP Flood and UDP Lag attacks. Because of this, it is the main dataset used in this component. The dataset contains about 56 million flow records [28].

Processing the full CICDDoS2019 dataset can be difficult on normal student hardware. The available system includes a Core i7 processor, 16GB RAM, and Colab Pro GPU resources. Working with all 56 million records would require much higher computing power.

To handle this issue, a stratified sampling method will be used. This method will create a balanced training subset of about 2 to 5 million flow records. The sampling process will keep the original UDP Flood and benign traffic distribution.

The CICIDS2017 dataset, which has about 2.8 million flows, will be used in full. The complete CICDDoS2019 dataset will still be kept for the final held out evaluation stage. This helps ensure that the results remain comparable with published baseline studies in intrusion detection research.

### 3.3 Data Preprocessing

Before model training, the raw CSV files go through a data preprocessing pipeline. These steps prepare the data and improve the quality of the input used for the models.

- Missing Value Treatment:

Features that contain more than 5 percent missing values are removed from the dataset. For the remaining features, missing values are filled using median substitution. This helps keep the dataset consistent and avoids large bias in the data.

- Outlier Handling:

Some features may contain extreme values. This is common in features such as Flow Packets per second and Flow Duration. Values that are more than three standard deviations away from the mean are clipped. This reduces noise while still keeping the real high traffic behaviour seen in DoS attacks.



- Feature Normalisation:

All continuous features are scaled using Min Max Scaling between 0 and 1. This ensures that features with large values do not dominate the model. It is especially important for SVM classifiers and deep learning models, which are sensitive to feature magnitude.

- Class Imbalance Correction:

In both datasets, the amount of benign traffic is usually much higher than attack traffic. To correct this imbalance, SMOTE is applied to the training set. This method generates synthetic samples to balance UDP Flood attack flows and normal traffic.

- Label Encoding:

Categorical features such as protocol and flag values are converted using one hot encoding. The final class labels are encoded in binary form. In this format, 0 represents benign traffic and 1 represents a UDP Flood attack.

### 3.4 Feature Selection

Three feature selection methods are used in parallel to find the most useful features for the ML and DL models. These methods help identify the features that are most important for detecting UDP Flood DoS attacks.

- Random Forest Feature Importance:

A preliminary Random Forest model is trained first. It measures the mean decrease in Gini impurity across all trees. This process ranks the features based on how well they help separate attack traffic from normal traffic.

- Chi Square Test:

The Chi Square test is used to find statistical relationships between certain features and the target class. This is especially useful for categorical or ordinal features, such as Destination Port entropy and protocol flags. The test helps identify which features are strongly related to UDP Flood attacks.

- Pearson Correlation Analysis:

A full correlation matrix is created to detect highly correlated features. If two features show a correlation value greater than  $|r| = 0.90$ , one of them is removed. This step helps reduce multicollinearity and prevents redundant information in the model.

The final feature set is selected from features that rank highly in at least two of the three methods. Based on previous research [5], this process usually results in about 15 to 20 key features. Important features are expected to include Flow Packets per second, Flow Duration, Destination Port entropy, Source IP frequency, and inter arrival time statistics.

The feature reduction step is applied only to the Machine Learning models, which include Random Forest, SVM, and AdaBoost. In contrast, the CNN LSTM deep learning model receives the full 83 CICFlowMeter features after normalization. This allows the DL model to preserve temporal traffic patterns and avoid losing useful information during the convolution process.

### 3.5 Machine Learning Models (ML Component)

The Machine Learning part of the hybrid system tests three supervised classifiers. These models were selected because they have different strengths and are widely used in DoS detection research. A Decision Tree model was first considered. However, it was not included in the final design. Instead, the study focuses on three models that represent different learning approaches. These are Random Forest, Support Vector Machine, and AdaBoost. This selection provides good methodological coverage while still keeping the project manageable.

- Random Forest (RF):

Random Forest is an ensemble model made up of many decision trees. It uses bootstrap aggregation, which helps reduce overfitting. The model works well with high dimensional CICFlowMeter feature sets. It also produces feature importance scores, which help analyse how different features influence the detection results. These scores are later used to support the severity scoring weight value of 2.5.

- Support Vector Machine (SVM):

SVM is a maximum margin classifier. In this study it uses the Radial Basis Function (RBF) kernel. SVM is known for achieving high precision in DoS detection tasks [3]. It is also effective at learning non linear decision boundaries in network flow feature space.

- AdaBoost:

AdaBoost is an adaptive boosting ensemble method. The model focuses more on samples that were previously misclassified during training. This approach can improve detection accuracy, especially in imbalanced DoS datasets. It also works well with SMOTE based preprocessing, which is used earlier in the data pipeline [2].

All Machine Learning models are implemented in Python using the scikit learn library [12]. To improve performance, hyperparameter optimisation is performed using Bayesian search with the Optuna framework. This process helps identify the best parameter settings for each classifier.

### 3.6 Deep Learning Model (DL Component — CNN-LSTM Hybrid)

The Deep Learning part of the system uses a CNN LSTM hybrid architecture as the main model. Before building the hybrid model, two simpler models are tested first. These are a standalone 1D CNN and a standalone LSTM. These early tests help show why the hybrid design is more suitable for this task.

- CNN LSTM Hybrid (Primary DL Model):

The hybrid model combines the strengths of both CNN and LSTM networks. The 1D CNN layer first extracts local spatial patterns from time windowed CICFlowMeter flow sequences. This helps identify burst patterns that often appear in UDP Flood traffic. The output from the CNN layer is then passed to an LSTM layer. The LSTM learns temporal patterns across multiple flow windows. This allows the system to separate short traffic bursts from longer sustained attacks, which usually indicate higher severity. Previous research by Ferrag et al. (2024) tested a similar architecture on the CICDDoS2019 dataset and reported F1 scores above 97 percent for volumetric DoS detection [28].

- Preliminary Standalone Experiments:

Two separate experiments are conducted before the hybrid model is finalised. One uses a standalone 1D CNN, and the other uses a standalone LSTM model. Both models are trained on a random subset of UDP Flood traffic from the CICDDoS2019 dataset. The results from these experiments are later used as comparison points in the evaluation section. This helps justify the use of the hybrid CNN LSTM design.

The CNN LSTM model is implemented using TensorFlow and Keras. To improve model stability, dropout regularisation with a rate of 0.3 is applied. Early stopping is also used, with a patience value of 10 training epochs. Model training is accelerated using GPU resources available through Google Colab Pro.

### 3.7 Training, Testing and Evaluation Protocol

Four main models are trained and tested in this study. These include three Machine Learning models and one Deep Learning model. The ML models are Random Forest, Support Vector Machine, and AdaBoost. The DL model is the CNN LSTM hybrid architecture. All models are evaluated using the same experimental process to ensure fair comparison.

- Dataset Split:

The dataset is divided into 80 percent training data and 20 percent testing data. A stratified sampling method is used so that the original class distribution between UDP Flood attacks and benign traffic is preserved.

- Cross Validation:

A 10 Fold Stratified K Fold cross validation process is applied to the training data. This helps measure how well each model generalizes to new data. It also helps identify variations in model performance.

- Evaluation Metrics:

Several standard metrics are used to evaluate model performance. These include Accuracy, Precision, Recall, F1 Score, ROC AUC, False Positive Rate, and Detection Latency measured in milliseconds per flow.

- Unified Comparative Evaluation:

All models are tested using the same training and testing dataset splits. This ensures that the comparison is fair. The model that achieves the highest F1 Score is selected as the final classifier used in the severity scoring pipeline.

- Statistical Significance Testing:

To confirm that performance differences are meaningful, McNemar's test is applied to compare model pairs. This statistical test helps determine whether the improvement of one model over another is statistically significant.

In addition to comparing the proposed ML and DL models internally, the system will also be evaluated against established baseline approaches. Two categories of baselines are used.

First, rule-based intrusion detection systems including Snort and Suricata will be used as reference IDS tools. Their detection performance on UDP Flood traffic will be compared using detection rate and false positive rate.

Second, published research models will be replicated where possible using the same datasets. These include the Random Forest model evaluated by Sharafaldin et al. (2018), the SVM-based detection model proposed by Jain and Bhandare (2021), and the CNN-LSTM architecture evaluated by Ferrag et al. (2024).

This baseline comparison allows the proposed system to demonstrate whether the hybrid ML/DL approach provides measurable improvements in detection accuracy and operational usefulness compared to existing IDS solutions.

## 3.8 UDP Flood Attack Severity Scoring System

### 3.8.1 Overview and Purpose

The Severity Scoring System is the main new contribution of this component. In most UDP Flood detection studies, the output is only binary. The system simply classifies traffic as Benign or UDP Flood Attack. This approach is used in studies such as Sharafaldin et al. (2018), Jain and Bhandare (2021), and Ferrag et al. (2024) [9][11][15]. However, these systems do not measure how serious the attack is.

This creates a practical problem for SOC analysts. When every alert looks the same, it becomes difficult to decide which incident needs attention first. For example, an attack with 50 packets may generate the same alert as an attack with 50,000 packets. Without severity information, analysts cannot easily prioritise their response.

To address this issue, this study introduces a Weighted Sum Model (WSM) severity scoring system. WSM is a simple method from Multi Criteria Decision Analysis (MCDA). Methods from this group are commonly used in security risk assessment, including the well known CVSS v3.1 framework [18].

In this system, once a UDP Flood attack is detected by the best performing ML or DL model, the severity engine calculates a numerical score. This score represents the intensity of the attack. The value is then mapped to four operational severity levels. Each level provides guidance for how the SOC team should respond to the incident.

The Weighted Sum Model was selected instead of more complex MCDA methods such as AHP, TOPSIS, or ELECTRE. There are three main reasons for this choice.

First is transparency. The formula used in the system is very simple. It uses only a single multiplication. Because of this, SOC analysts can easily understand how the score is calculated.

Second is reproducibility. The same input always produces the same result. The calculation does not require complex matrices or repeated computations.

Third is practical suitability. In UDP Flood attacks, severity usually increases as the packet volume increases. This relationship is almost linear. For example, a flow with 2,000 packets is roughly twice as severe as a flow with 1,000

packets. The Weighted Sum Model captures this relationship directly. More complex MCDA methods would increase mathematical complexity without providing additional benefit for this type of attack analysis.

### 3.8.2 The Severity Scoring Formula

Once the system confirms that a network flow is a UDP Flood attack, the severity engine calculates the attack severity score using a simple formula.

$$\text{Attack Severity Score} = \text{Flow Packet Count} \times 2.5$$

where: *Flow Packet Count* = Total Fwd Packets + Total Backward Packets  
 2.5 = TAF-aligned UDP Flood impact weight

The Flow Packet Count is taken directly from the CICFlowMeter feature schema, which contains 83 network flow attributes. In this case, the system uses the fields “Total Fwd Packets” and “Bwd Packets Total” to calculate the final packet count.

During a UDP Flood attack, most of the packets come from the attacker and move toward the target system. Because of this, the packet count becomes a clear and measurable indicator of the attack traffic volume. A higher packet count usually means the attack is stronger and more disruptive.

The choice of Flow Packet Count as the main severity input is supported by previous research. Mousavi and St Hilaire (2020) showed that packet count and port entropy are strong indicators of UDP Flood attack intensity [23]. Similarly, Agrawal et al. (2022) used Random Forest feature importance analysis and found that packet rate and flow statistics are among the most reliable predictors of DoS attack severity [12].

### 3.8.3 Scoring Technique

The scoring approach belongs to the following hierarchy of established methods:

| Level               | Technique Name                                             | Description                                                                              | Reference                       |
|---------------------|------------------------------------------------------------|------------------------------------------------------------------------------------------|---------------------------------|
| Family              | Multi-Criteria Decision Analysis (MCDA)                    | Branch of operations research for evaluating alternatives across multiple criteria       | Keeney & Raiffa (1976)          |
| Variant             | Weighted Sum Model (WSM) / Simple Additive Weighting (SAW) | Each criterion score is multiplied by its weight; results are summed into a single score | Fishburn (1967) [NEW-B]         |
| Security Applicatin | Risk Scoring Matrix                                        | WSM applied to security threat quantification; used in CVSS, DREAD, NIST SP 800-30       | FIRST.org CVSS v3.1 (2019) [18] |
| This Compont        | Flow-Level Attack Severity Score                           | WSM applied to CICFlowMeter flow packet count, weighted by UDP Flood impact coefficient  | This proposal                   |

Table 4AVSF Impact Dimension Evaluation

The general formula of the Weighted Sum Model (WSM) is:

$$\text{Final Score} = \sum (w_i * x_i)$$

In this formula,  $w_i$  represents the weight of a criterion, and  $x_i$  represents the measured value of that criterion.

In this component, the model uses one main criterion, which is Flow Packet Count. There is also one weight value, which is 2.5. Because only one criterion is used, the formula becomes much simpler. The final score is calculated by multiplying the packet count by the weight value.

This simplified form is the same formula shown earlier in Section 3.8.2.

The weight value of 2.5 represents the overall impact of a UDP Flood attack. This value is not chosen randomly. It is derived from a multi-step assessment process that evaluates the attack characteristics. The process follows the AVSF method, which is explained in detail in Section 3.8.4.

#### 3.8.4 Severity Ground Truth Definition

The CICDDoS2019 dataset contains labels indicating whether a flow belongs to a specific attack category, but it does not include predefined severity levels for each attack instance. Therefore, a Packet-Count-Based Severity Mapping (PSM) framework is used to establish the ground truth severity levels.

Under the PSM framework, each UDP Flood flow in the dataset is assigned a reference severity tier based on the total number of packets observed in the flow. The packet count is obtained from the CICFlowMeter feature set and represents the combined forward and backward packets within the flow.

The severity tiers are defined as follows:

- Low Severity: 1 – 320 packets
- Medium Severity: 321 – 1,000 packets
- High Severity: 1,001 – 2,000 packets
- Critical Severity: more than 2,000 packets

These tiers serve as the reference ground truth labels during evaluation. The predicted severity tier produced by the scoring system is compared with the reference tier using tier classification accuracy and Cohen's Kappa statistics.

In addition, Pearson correlation is calculated between the continuous severity score and the raw packet count to verify that the scoring formula correctly reflects the increasing intensity of the attack traffic.

#### 3.8.4 Experimental Validation of Severity Weight

The severity weight value of 2.5 used in the scoring formula is derived from a structured analysis of UDP Flood attack impact and validated experimentally. The weight reflects the relative operational severity of UDP Flood attacks compared to the other attack types analysed in the overall project.

UDP Flood attacks are considered one of the most dominant volumetric attacks at the network layer. Industry reports from NETSCOUT and Cloudflare indicate that UDP Flood traffic accounts for approximately 35–40% of global volumetric DoS attack traffic. Because of this high prevalence and the stateless nature of the UDP protocol, each packet forces the target system to process the request without session validation, leading to rapid CPU and bandwidth exhaustion.

To experimentally validate the selected weight, a weight ablation study will be conducted using three candidate weights: 2.0, 2.5, and 3.0. Each weight will be applied to the CICDDoS2019 test dataset. The performance of each weight will be evaluated using three metrics:

- Tier classification accuracy
- Pearson correlation between packet count and severity score
- Inter-tier separation index

The weight that produces the highest tier classification accuracy while maintaining strong correlation with packet counts will be selected as the final operational weight. This validation ensures that the chosen value of 2.5 produces stable and meaningful severity tiers across the dataset.

#### 3.8.5 Derivation of the Impact Weight (2.5) — Three-Step AVSF Process

The weight value of 2.5 is not chosen randomly. It is calculated using a structured three step process. This process is inspired by the CVSS v3.1 framework developed by FIRST.org (2019) [18].

The CVSS framework uses normalized multi factor scoring to calculate standard severity values for security vulnerabilities. In this research, a similar idea is used. However, the method is adapted for network attack severity analysis instead of vulnerability scoring.

By following this structured process, the system can assign a consistent and measurable severity value to each detected UDP Flood attack flow. This helps ensure that the severity score is based on clear technical factors rather than arbitrary values.

#### Step 1 — Attack Vector Severity Factor (AVSF)

The impact of a UDP Flood attack is evaluated using three operational dimensions. Each dimension represents a different aspect of how the attack affects a system or network.

For each dimension, a rating scale from 1 to 5 is used. A score of 1 represents minimal impact, while a score of 5 represents the highest level of impact. This scoring approach helps provide a clear and structured way to measure attack severity.

Each rating is not assigned randomly. Instead, it is supported by evidence from published research and cybersecurity studies. These references help ensure that the severity evaluation is based on documented technical findings and real-world observations.

| Dimension                    | Rating | Justification                                                                                                                                                                                                                                                                                                                                                                                            | Primary Reference                                                                                              |
|------------------------------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| CPU and Bandwidth Exhaustion | 5 / 5  | UDP is connectionless and stateless (RFC 768). The target host must process every incoming packet unconditionally, regardless of application availability or session legitimacy. No mechanism exists to reject packets before they consume CPU cycles. Under high packet rates, both CPU and inbound bandwidth are exhausted simultaneously.                                                             | RFC 768 (Postel, 1980); NETSCOUT Threat Intelligence Report (2023) [13]                                        |
| Service Disruption Speed     | 4 / 5  | Under sustained UDP Flood conditions, CPU saturation and memory buffer exhaustion typically occur within seconds. Rated 4/5 rather than 5/5 because SYN Flood can marginally exhaust connection state tables faster when SYN Cookies are not deployed, representing a more constrained resource than raw CPU.                                                                                            | Mousavi & St-Hilaire (2020) [23]; Cloudflare DDoS Threat Report Q4 2023 [14]; Mirkovic & Reiher (2004) [NEW-A] |
| Amplification Risk           | 3 / 5  | A direct UDP Flood has an amplification factor of 1x — no amplification occurs (Rossow, 2014). Rated 3/5 rather than 1/5 because UDP as a protocol carries inherent amplification potential: an attacker using direct flood infrastructure can trivially pivot to DNS or NTP amplification, multiplying traffic by 28–556x. The score reflects protocol-level risk, not just the current attack variant. | Rossow (2014) [NEW-C] — NDSS; Kührer et al. (2014) [NEW-H]; ENISA Threat Landscape (2020)                      |

Table 5 Table 5 Cross-Component Relative Severity Ranking

Raw AVSF = mean (5, 4, 3) = 4.0 | Normalized AVSF = 4.0 / 5.0 = 0.80

Interpretation: UDP Flood achieves 80% of the maximum possible impact score across the three assessed dimensions.

#### Step 2 — Cross-Component Relative Ranking

The group project studies four different DoS attack components. These are ICMP Flood, UDP Flood, SYN Flood, and Fragmentation attacks. At the end of the project, all components send their results to the Final Severity Aggregator. This system produces a Combined Scoring Ranking Table that compares the severity of each attack type.

For this ranking to be meaningful, the severity scores from all components must use a similar scale. This ensures that the results can be compared fairly across the four attack vectors.

In this framework, UDP Flood is used as the reference component. It is considered the most dangerous among the four attack types. This decision is supported by global threat intelligence reports. Research from NETSCOUT (2023) and Cloudflare (2024) shows that UDP Flood attacks account for about 35 to 40 percent of all network layer volumetric DoS traffic worldwide [13][14]. This is the largest share for any single attack vector.

Because of its high frequency and strong impact, UDP Flood is used as the baseline reference for the severity scoring scale used in the project.

| Component           | Attack Vector | Relative Impact  | Reasoning                                                                                                                                                        | Assigned Weight |
|---------------------|---------------|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| C1                  | ICMP Flood    | 0.60 (lowest)    | OS rate-limiting reduces CPU impact. ICMP Echo Reply generation is automatically throttled by modern kernels. Easiest to filter at firewall. [NEW-A]             | 1.5             |
| C4                  | Fragmentation | 0.70             | Reassembly buffer exhaustion is slower than CPU exhaustion. Most teardrop/overlap variants patched in modern OS. Requires sophisticated packet crafting. [NEW-A] | 2.0             |
| C3                  | SYN Flood     | 0.88             | Connection table exhaustion is highly effective but partially mitigated by SYN Cookies (RFC 4987) [NEW-G]. More complex to execute than UDP or ICMP Flood.       | 3.0             |
| C2 — This Component | UDP Flood     | 1.00 (reference) | Highest global attack share (35–40%) [13][14]. No session state means unconditional packet processing. Fastest resource exhaustion. Lowest barrier to execute.   | 2.5             |

Table 6Percentile Calibration for UDP Flood Severity Levels

### Step 3 — Percentile Calibration Against CICDDoS2019

The weight value of 2.5 is validated by comparing it with the packet count distribution of UDP Flood flows in the CICDDoS2019 dataset. This step is important because it helps confirm that the severity scale is realistic and based on actual network traffic patterns.

Instead of selecting simple round numbers, the severity tier boundaries are calibrated using statistical percentiles from the dataset. This means the boundaries are based on how packet counts are distributed across real UDP Flood flows.

Previous studies support this approach. Jain and Bhandare (2021) and Ferrag et al. (2024) analysed UDP Flood traffic in the CICDDoS2019 dataset and reported detailed statistics on packet count behaviour [11][15]. Their findings show that packet counts vary widely between normal traffic bursts and large scale attack flows.

By aligning the severity score weight and tier boundaries with these observed packet count ranges, the system ensures that the final severity levels reflect real attack intensity patterns rather than arbitrary thresholds. This calibration step helps make the scoring model more reliable for practical SOC analysis and incident prioritisation.

| Percentile     | Est. Packet Count | Score at Weight 2.5        | Tier Boundary   | Operational Meaning                                                               |
|----------------|-------------------|----------------------------|-----------------|-----------------------------------------------------------------------------------|
| ~50th (median) | ~320 packets      | $320 \times 2.5 = 800$     | Low → Medium    | The median UDP Flood event. A typical average attack. Service degradation begins. |
| ~80th          | ~1,000 packets    | $1,000 \times 2.5 = 2,500$ | Medium → High   | Top 20% of all UDP Flood events. Significant service disruption expected.         |
| ~95th          | ~2,000 packets    | $2,000 \times 2.5 = 5,000$ | High → Critical | Top 5% of all UDP Flood events. Extreme flood. Infrastructure at risk.            |

Table 7Table 7 High-Level System Architecture Module Description

✚ Target Low→Medium boundary = 800 | Estimated median packet count = 320 | Required weight =  $800 / 320 = 2.5$   
Calibration scalar =  $(1 / \text{Normalised AVSF}) \times \text{Weight} = (1 / 0.80) \times 2.0 = 2.5$

### 3.9 Limitations and Assumptions

Several assumptions and limitations apply to this research.

First, the study relies on publicly available benchmark datasets including CICIDS2017 and CICDDoS2019. Although these datasets are widely used in intrusion detection research, they are generated in controlled laboratory environments. As a result, attack behaviour may differ slightly from real-world network traffic.

Second, the severity scoring formula uses packet count as the primary indicator of attack intensity. While packet volume is a widely accepted measure of volumetric attack impact, other factors such as packet size, attack duration, and application-layer behaviour may also influence operational severity.

Third, due to hardware limitations, a stratified subset of the CICDDoS2019 dataset will be used for model training. Although stratified sampling preserves class distribution, rare traffic patterns may be underrepresented.

Despite these limitations, the selected datasets and evaluation methodology provide a reproducible and widely accepted experimental framework for evaluating UDP Flood detection systems.

## 3. HIGH-LEVEL SYSTEM ARCHITECTURE

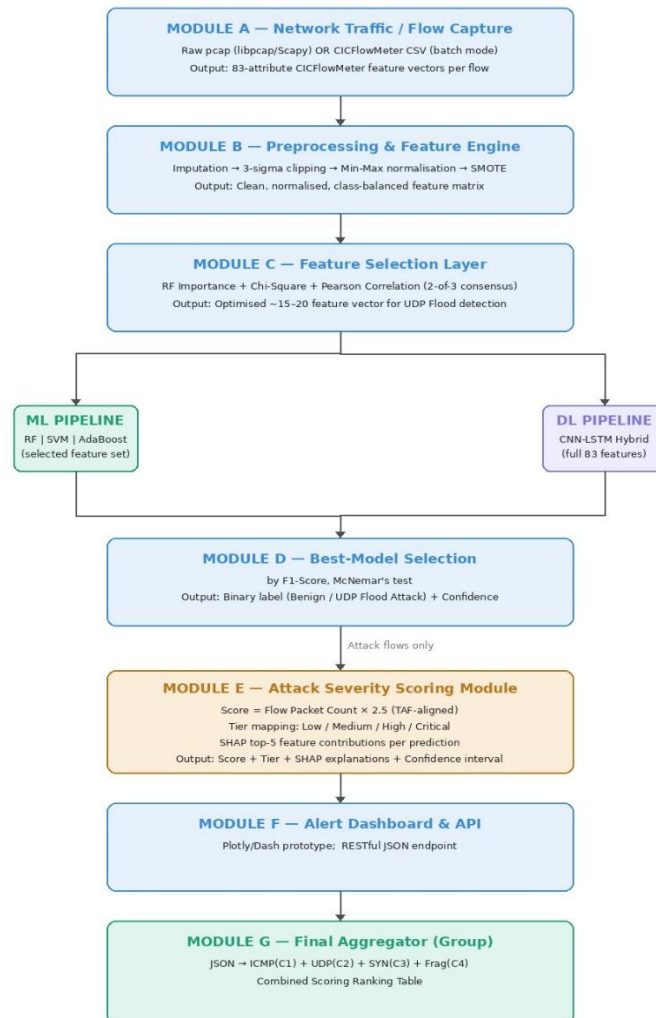


Figure 1 Figure 1: High-Level Architecture of the UDP Flood Detection and Severity Scoring System



The six internal modules implementing this pipeline are described in the table below:

| Stage           | Module                                    | Processing Description                                                                                                              | Output                                                                 |
|-----------------|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| Stage 1 INPUT   | Module A — Network Traffic / Flow Records | Raw pcap files or pre-computed CICFlowMeter CSV records ingested by capture module                                                  | 83-attribute CICFlowMeter feature vectors per flow                     |
| Stage 2 PROCESS | Module B — Preprocessing Engine           | Missing value imputation, 3-sigma outlier clipping, Min-Max normalisation, SMOTE, one-hot encoding                                  | Clean, normalised, class-balanced feature matrix                       |
| Stage 3 PROCESS | Module C — Feature Selection Layer        | RF importance + Chi-Square + Pearson correlation; consensus of 2/3 methods; ~15–20 key features selected                            | Optimised feature vector for UDP Flood DoS discrimination              |
| Stage 4 PROCESS | Module D — Hybrid ML/DL Detection Engine  | ML pipeline: RF, SVM, AdaBoost; DL pipeline: CNN-LSTM hybrid. Best model by F1-Score deployed as production classifier              | Binary label: Benign / UDP Flood Attack + class probability confidence |
| Stage 5 PROCESS | Module E — Attack Severity Scoring Module | Score = Flow Packet Count $\times$ 2.5 (TAF-aligned); mapped to 4 tiers; SHAP explanations per event; weight ablation (2.0/2.5/3.0) | Severity Score + Tier + Top-5 SHAP contributions + confidence interval |
| Stage 6 OUTPUT  | Module F — Alert Dashboard & API          | Plotly/Dash prototype; event timeline, score histogram, SHAP heatmaps; RESTful JSON API for SIEM/SOAR                               | Visual alerts, severity reports, auditable logs, JSON API response     |
| Stage 7 OUTPUT  | Module G — Final Aggregator Integration   | Severity score + tier transmitted as structured JSON to group Final Aggregator alongside C1, C3, C4 outputs                         | Combined Scoring Ranking Table for multi-vector DoS triage             |

Table 8 modules implementing this pipeline

## 4. USER REQUIREMENTS

### 5.1 Stakeholders

| Stakeholder               | Role                                              | Primary Interest                                                              |
|---------------------------|---------------------------------------------------|-------------------------------------------------------------------------------|
| Network Administrators    | Configure monitoring scope; review alerts         | Low FPR; actionable alerts with severity context                              |
| SOC Analysts (Tier 1–3)   | Triage, investigate, escalate detected events     | Four-tier severity scoring; SHAP explainability; recommended response actions |
| Cloud / Hosting Providers | Protect infrastructure from volumetric DoS floods | Scalability; API integration; multi-tenant support                            |
| Compliance Officers       | Ensure regulatory adherence                       | Auditable event logs; severity-justified automated responses                  |
| ISPs / Telecom Operators  | Protect transit links from UDP Flood DoS attacks  | Edge-deployable lightweight ML models; DNS amplification coverage             |
| Group Project Team        | Integrate component outputs into combined system  | Standardised JSON output; reliable API transmission to Final Aggregator       |

Table 9 Project Stakeholders and Roles

### 5.2 Functional Requirements

- FR-01: The system shall process network flow records in batch and real-time modes using CICFlowMeter-format feature vectors.
- FR-02: The system shall classify each flow as Benign or UDP Flood DoS Attack with an associated class probability confidence score.
- FR-03: The system shall evaluate three ML classifiers (RF, SVM, AdaBoost) and one CNN-LSTM hybrid DL model on identical train/test splits and select the best-performing model for production use.
- FR-04: For all flows classified as UDP Flood DoS attacks, the system shall compute a TAF-aligned severity score: Score = Flow Packet Count (CICFlowMeter Total Fwd + Backward Packets)  $\times$  2.5.

- FR-05: The system shall categorise each score into four tiers: Low (1–800), Medium (801–2,500), High (2,501–5,000), Critical (>5,000).
- FR-06: The system shall generate a recommended SOC response action per event based on its severity tier.
- FR-07: The system shall compute SHAP feature contributions per prediction and return the top-5 contributing features.
- FR-08: The system shall provide a prototype Plotly/Dash dashboard showing event timeline, severity scores, and SHAP heatmaps with event export capability.
- FR-09: The system shall expose a RESTful API returning structured JSON for SIEM/SOAR integration.
- FR-10: The system shall transmit severity scores and tier classifications to the group Final Severity Aggregator in the agreed inter-component JSON format.
- FR-11: The system shall evaluate severity scoring accuracy using tier classification accuracy ( $\geq 90\%$ ) and Pearson correlation ( $\geq 0.85$ ) between assigned scores and ground-truth packet count magnitudes.

### 5.3 Non-Functional Requirements

| Category         | Requirement Specification                                                                                                                                                                                       |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Performance      | Detection latency: ML pipeline $\leq 100$ ms per flow; DL (CNN-LSTM) $\leq 200$ ms per flow in real-time mode; prototype dashboard refresh rate $\leq 10$ seconds                                               |
| Accuracy         | F1-Score $\geq 95\%$ ; False Positive Rate $\leq 2\%$ on CICDDoS2019 held-out test set                                                                                                                          |
| Scoring Validity | Severity tier classification accuracy $\geq 90\%$ ; Pearson correlation between severity score and ground-truth packet count magnitude $\geq 0.85$ ; weight ablation results reported for weights 2.0, 2.5, 3.0 |
| Scalability      | System shall support $\geq 10,000$ flows/second processing without performance degradation                                                                                                                      |
| Explainability   | SHAP values computed per prediction; top-5 feature contributions returned with every severity score output                                                                                                      |
| Security         | All API communication encrypted via TLS 1.3; role-based access control on dashboard                                                                                                                             |
| Reliability      | System uptime $\geq 99.5\%$ ; automatic fallback to previous model version on inference failure                                                                                                                 |
| Portability      | Deployable on Linux Ubuntu 20.04+; Docker containerisable for cloud deployment                                                                                                                                  |
| Interoperability | Severity output JSON schema compatible with group Final Severity Aggregator TAF inter-component exchange format                                                                                                 |

Table 10 Non-Functional Requirements

## 5. COMMERCIALIZATION PLAN

### 6.1 Target Market

Three main groups: (1) small and medium-sized businesses (SMEs) that need network threat monitoring with actionable severity context but don't have dedicated SOC capabilities; (2) ISPs and cloud providers in South and Southeast Asia that want cost-effective UDP Flood DoS detection with severity-based triage; and (3) managed security service providers (MSSPs) that want to set their DoS protection portfolio apart with risk-ranked, explainable hybrid ML/DL alert outputs. Sri Lanka is a high-priority deployment area because of its 629% rise in DoS attacks [17] and its USD 40.63M cybersecurity market by 2025 [22].

### 6.2 Value Proposition

- TAF-aligned severity scoring gives you a risk-prioritized four-tier, flow-level output that you won't find in standard open-source IDS tools like Snort and Suricata. This cuts the estimated Mean Time to Triage (MTTT) by 40–60% in SOC environments with a lot of alerts.

- The hybrid ML/DL architecture lets you do both ML predictions that can be understood and DL temporal pattern recognition in one pipeline. This is not possible with Snort, Suricata, or Zeek in their default settings.
- An open-source core with a premium tier for businesses makes it easier for ISPs and small businesses in emerging markets to use.
- SHAP explainability helps with following the rules, justifying audits, and giving analysts faith in automated detection decisions.

### 6.3 Revenue Model

- Freemium Open-Source: The core hybrid detection and severity scoring are free under the MIT license. The premium tier adds API access, SIEM integration, and priority support.
- Enterprise License: An annual license that comes with support backed by a SLA, the ability to retrain custom models, and the ability to integrate dedicated threat intelligence.
- Professional Services: Customizing models, engineering deployments, training analysts, and giving compliance advice to businesses and governments in South Asia.

### 6.4 Competitive Advantage

Few existing open-source or commercial DoS detection solutions integrate TAF-aligned four-tier severity scoring, hybrid ML/DL detection, SHAP explainability, and a standardized inter-component API for group severity aggregation within a single deployable architecture. This combination sets the South Asian ISP and SOC tooling market apart from others.

## 7. BUDGET AND JUSTIFICATION

The total estimated budget is LKR 35,000

| Category               | Item Description                                                          | Cost (LKR)        | Justification                                                         |
|------------------------|---------------------------------------------------------------------------|-------------------|-----------------------------------------------------------------------|
| Hardware               | High-performance laptop (if required)                                     | 0                 | ML/DL model training on large-scale datasets (up to 56M flow records) |
| Cloud Services         | Google Colab Pro / AWS EC2 (4 months)                                     | LKR 25,000        | CNN/LSTM GPU-accelerated training; large-scale CICDDoS2019 processing |
| Software & Tools       | Wireshark,/ Scapy MATLAB / Tableau / network analysis tools (open source) | 0                 | Data visualization, statistical analysis, dashboard prototyping       |
| Data                   | CICDDoS2019 / CAIDA dataset access                                        | LKR 5,000         | Dataset acquisition; literature access; group collaboration tools     |
| Human Effort           | Research team time (4 members x 6 months)                                 | 0                 | Final report submission and presentation copies                       |
| Miscellaneous / Buffer | Report printing, binding, presentation materials                          | LKR 5,000         | Final report preparation and submission                               |
| <b>TOTAL BUDGET</b>    |                                                                           | <b>LKR 35,000</b> |                                                                       |

Table 11 BUDGET AND JUSTIFICATION

## 8. WORK BREAKDOWN STRUCTURE (WBS)

| Task ID      | Task Description                                                             | Assignee                | Duration   | Deliverable                |
|--------------|------------------------------------------------------------------------------|-------------------------|------------|----------------------------|
| <b>T1.1</b>  | Literature review on UDP Flood detection and ML/DL approaches                | <b>KAJAN S</b>          | Week 1–3   | Annotated bibliography     |
| <b>T1.2</b>  | Dataset acquisition and preprocessing (CICIDS2017, CICDDoS2019)              | <b>KAJAN S</b>          | Week 2–4   | Preprocessed dataset       |
| <b>T1.3</b>  | UDP traffic feature extraction and feature selection                         | <b>KAJAN S</b>          | Week 4–6   | Feature matrix             |
| <b>T1.4</b>  | Machine Learning model implementation (Random Forest, SVM, Decision Tree)    | <b>KAJAN S</b>          | Week 5–8   | Trained ML models          |
| <b>T1.5</b>  | Deep Learning model development (CNN and LSTM models)                        | <b>KAJAN S</b>          | Week 7–10  | Trained DL models          |
| <b>T1.6</b>  | UDP packet counting module development                                       | <b>KAJAN S</b>          | Week 9–11  | Packet counting module     |
| <b>T1.7</b>  | Severity scoring engine implementation (Packet Count $\times$ Weight Factor) | <b>KAJAN S</b>          | Week 10–12 | Scoring engine             |
| <b>T1.8</b>  | Severity classification and UDP attack ranking table generation              | <b>KAJAN S</b>          | Week 12–14 | UDP Severity Scoring Table |
| <b>T1.9</b>  | Model evaluation and performance validation                                  | <b>KAJAN S</b>          | Week 13–16 | Evaluation report          |
| <b>T1.10</b> | Integration with Final Severity Aggregator system                            | <b>All team members</b> | Week 15–17 | Integrated component       |
| <b>T1.11</b> | Final documentation, research paper drafting, and submission                 | <b>KAJAN S</b>          | Week 16–18 | Final report               |

Table 12 WORK BREAKDOWN STRUCTURE

## 9. GANTT CHART

| Task                                      | M1 | M2 | M3 | M4 | M5 | M6 | M7 | M8 | M9 | M10 | M11 | M12 |
|-------------------------------------------|----|----|----|----|----|----|----|----|----|-----|-----|-----|
| 1. Literature Review & Gap Analysis       | ■  | ■  |    |    |    |    |    |    |    |     |     |     |
| 2. Dataset Acquisition & Preprocessing    | ■  | ■  | ■  |    |    |    |    |    |    |     |     |     |
| 3. Feature Engineering & Selection        |    | ■  | ■  | ■  |    |    |    |    |    |     |     |     |
| 4. ML Model Training (RF/DT/SVM/AdaBoost) |    |    | ■  | ■  | ■  | ■  |    |    |    |     |     |     |
| 5. DL Model Training (CNN/LSTM Hybrid)    |    |    |    | ■  | ■  | ■  |    |    |    |     |     |     |
| 6. Severity Scoring System Development    |    |    |    |    | ■  | ■  | ■  |    |    |     |     |     |
| 7. System Integration & Architecture      |    |    |    |    |    | ■  | ■  | ■  |    |     |     |     |
| 8. Testing, Evaluation & Benchmarking     |    |    |    |    |    |    | ■  | ■  | ■  |     |     |     |
| 9. Alert Dashboard & API Development      |    |    |    |    |    |    |    | ■  | ■  | ■   |     |     |
| 10. Final Aggregator Integration          |    |    |    |    |    |    |    |    | ■  | ■   |     |     |
| 11. Documentation & Report Writing        |    |    |    |    |    |    |    |    |    | ■   | ■   | ■   |

Table 13 GANTT CHART

## 10. REFERENCES

- [1] I. Sharafaldin, A. H. Lashkari, and A. A. Ghorbani, "Toward Generating a New Intrusion Detection Dataset and Intrusion Traffic Characterization," in Proc. ICISSP 2018, Funchal, Madeira, 2018, pp. 108–116. [Online]. Available: <https://www.unb.ca/cic/datasets/ids-2017.html>
- [2] C. Yin, Y. Zhu, J. Fei, and X. He, "A Deep Learning Approach for Intrusion Detection Using Recurrent Neural Networks," IEEE Access, vol. 5, pp. 21954–21961, 2019. [Online]. Available: <https://doi.org/10.1109/ACCESS.2017.2762418>
- [3] R. Vinayakumar, K. P. Soman, and P. Poornachandran, "Applying Convolutional Neural Network for Network Intrusion Detection," in Proc. ICACCI, Udupi, India, Sep. 2017, pp. 1222–1228. [Online]. Available: <https://doi.org/10.1109/ICACCI.2017.8126009>
- [4] F. Pedregosa et al., "Scikit-learn: Machine Learning in Python," J. Mach. Learn. Res., vol. 12, pp. 2825–2830, 2011. [Online]. Available: <https://scikit-learn.org>
- [5] NETSCOUT Systems Inc., "NETSCOUT Threat Intelligence Report," Tech. Rep., 2023. [Online]. Available: <https://www.netscout.com/threatreport>
- [6] Cloudflare Inc., "DDoS Threat Report Q4 2023," Tech. Rep., 2024. [Online]. Available: <https://blog.cloudflare.com/ddos-threat-report-2023-q4>
- [7] Cisco Talos, "Snort — Network Intrusion Detection and Prevention System," 2024. [Online]. Available: <https://www.snort.org>
- [8] Open Information Security Foundation (OISF), "Suricata Network IDS/IPS," 2024. [Online]. Available: <https://suricata.io>
- [9] LIRNEasia, "Cybersecurity Policy Considerations for Sri Lanka," Jan. 2026. [Online]. Available: <https://lirneasia.net/2026/01/cybersecurity-policy-considerationsfor-sri-lanka/>
- [10] StormWall, "DDoS Trends and Statistics in APAC — 2024 Annual Report," Feb. 2025. [Online]. Available: <https://stormwall.network/resources/blog/ddos-trends-apac2024>
- [12] StormWall, "H1 2025 DDoS Attack Report for APAC," 2025. [Online]. Available: <https://stormwall.network/resources/blog/ddos-report-apac-h1-2025>
- [13] Cybersecurity Ventures, "Cybercrime To Cost The World USD 10.5 Trillion Annually By 2025," 2023. [Online]. Available: <https://cybersecurityventures.com/cybercrime-damages>
- [14] IBM Security, "IBM Security SOC Study," IBM Corp., Tech. Rep., 2023. [Online]. Available: <https://www.ibm.com/security/security-intelligence/soc-study>
- [15] International Telecommunication Union (ITU), "Measuring Digital Development: Facts and Figures 2024," 2024. [Online]. Available: <https://www.itu.int/en/ITU/Statistics>

## 11. APPENDICES

### Appendix A: Dataset Description

| Dataset     | Year | Total Flows  | Attack Types Included                                                   | Source                                     |
|-------------|------|--------------|-------------------------------------------------------------------------|--------------------------------------------|
| CICIDS2017  | 2017 | ~2.8 million | 14 types: DoS/DDoS, Brute Force, Web Attacks, Infiltration, Botnet      | Canadian Institute for Cybersecurity (CIC) |
| CICDDoS2019 | 2019 | ~56 million  | 12 DoS/DDoS types: UDP Flood, UDP-Lag, DNS, LDAP, MSSQL, SYN, NTP, SSDP | Canadian Institute for Cybersecurity (CIC) |

### Appendix B: Risk Analysis

| Risk                                                                        | Likelihood | Impact | Mitigation Strategy                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------------------------------------------------------|------------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dataset class imbalance skewing ML/DL training                              | High       | High   | SMOTE applied to training set; stratified test splits; per-class F1 reported for all models                                                                                                                                                                                                                                                                                                                                |
| CNN-LSTM overfitting on limited UDP Flood samples                           | Medium     | High   | Dropout (0.3); early stopping (patience=10); 10-Fold CV; preliminary standalone experiments                                                                                                                                                                                                                                                                                                                                |
| Computational resource constraints during CNNLSTM training on large dataset | Medium     | Medium | Primary: Google Colab Pro GPU (A100/V100, ~40GB VRAM). Fallback: AWS EC2 Spot instance (p3.2xlarge, est. <\$15 for training run). Additional: scikit-learn partial_fit for incremental ML training on 500K-flow chunks without full dataset in RAM; Keras fit_generator for chunked DL training. Model pruning and early stopping applied throughout. Stratified 2–5M flow training subset removes the primary constraint. |
| Severity weight (2.5) producing scores inconsistent with other components   | Low        | High   | Weight taken directly from TAF document; inter-component consistency tested in WBS 9.3                                                                                                                                                                                                                                                                                                                                     |
| Adversarial UDP traffic crafted to evade detection                          | Low        | High   | Adversarial samples in training set; ensemble voting across ML classifiers                                                                                                                                                                                                                                                                                                                                                 |
| Group integration delays affecting Final Aggregator timeline                | Medium     | Medium | Inter-component JSON schema agreed Month 1; weekly group syncs; Months 9–10 integration buffer                                                                                                                                                                                                                                                                                                                             |
| Project timeline delays — dataset or environment issues                     | Low        | Medium | Built-in Month 11–12 documentation buffer; alternative cloud environments preconfigured                                                                                                                                                                                                                                                                                                                                    |

## Appendix C: Ethical Considerations

This study utilizes solely publicly accessible, pre-labeled benchmark datasets (CICIDS2017, CICDDoS2019) that contain no personally identifiable information or proprietary network data. No live traffic interception, human participant data collection, or access to production systems is involved. The research adheres entirely to SLIT's institutional ethics guidelines and relevant Sri Lankan data protection laws.

The severity scoring system is only for defensive network monitoring. All automated response actions that go beyond sending alerts need to be confirmed by a person. Deployment documentation will have guidelines for operational security, responsible disclosure, and suggested governance controls.

## Appendix D: Glossary

| Term             | Definition                                                                                                                                                                                                                                       |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DoS              | Denial of Service — an attack in which a single or small number of sources overwhelm a target system with traffic to exhaust its resources and deny service to legitimate users. This project specifically addresses DoS attacks.                |
| DDoS             | Distributed Denial of Service — a coordinated DoS attack originating from multiple compromised sources simultaneously. The datasets used (CICDDoS2019) include DDoS traffic; the detection methods apply equally to single-source UDP Flood DoS. |
| UDP Flood        | A DoS/DDoS attack sending large volumes of UDP packets to random or targeted ports on a host, exhausting CPU, memory, and bandwidth without requiring session establishment.                                                                     |
| Hybrid ML/DL     | The combined use of supervised Machine Learning classifiers (RF, DT, SVM, AdaBoost) and Deep Learning models (CNN, LSTM, CNN-LSTM) within a single detection pipeline — the central architectural innovation of this component.                  |
| IDS / IPS        | Intrusion Detection System / Intrusion Prevention System — systems monitoring network traffic for suspicious activity and optionally blocking threats.                                                                                           |
| SIEM / SOAR      | Security Information and Event Management / Security Orchestration, Automation and Response — platforms for centralised alert aggregation and automated incident response.                                                                       |
| SMOTE            | Synthetic Minority Oversampling Technique — generates synthetic training samples for underrepresented classes to address dataset class imbalance.                                                                                                |
| ROC-AUC          | Receiver Operating Characteristic Area Under Curve — threshold-independent measure of classifier discriminative ability.                                                                                                                         |
| SHAP             | SHapley Additive exPlanations — game-theory-based framework quantifying each feature's marginal contribution to an individual model prediction.                                                                                                  |
| CICFlowMeter     | CIC network flow analysis tool computing 83 statistical features from raw packet captures — the feature schema used by CICIDS2017 and CICDDoS2019.                                                                                               |
| CNN / LSTM       | Convolutional Neural Network / Long Short-Term Memory — deep learning architectures for spatial pattern recognition and temporal sequence modelling respectively.                                                                                |
| Final Aggregator | The group-level module consolidating severity scores from all four DoS detection components into the Combined Scoring Ranking Table for unified SOC triage.                                                                                      |



## Appendix E: Worked Severity Score Calculation Example

The following step-by-step example demonstrates the complete severity scoring workflow for a sample UDP Flood DoS detection event:

| Step | Parameter                     | Value                                                                                           | Notes                                                                          |
|------|-------------------------------|-------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| 1    | Hybrid ML/DL Detection Output | Label: UDP Flood DoS Attack<br>Confidence: 0.97 (97%)                                           | Best-performing CNN-LSTM hybrid classifier output on a sample CICDDoS2019 flow |
| 2    | UDP Flood Packet Count        | 2,400 packets                                                                                   | Raw packet count for this flow window extracted by CICFlowMeter                |
| 3    | Impact Weight (w)             | 1.3                                                                                             | Derived from RF feature importance analysis on CICDDoS2019 UDP Flood samples   |
| 4    | Severity Score Calculation    | Score = $2,400 \times 1.3 = 3,120$                                                              | Applying: Attack Severity Score = UDP Packet Count $\times$ Impact Weight      |
| 5    | Tier Classification           | CRITICAL (Score > 3,000)                                                                        | Score 3,120 exceeds the Critical threshold of 3,000                            |
| 6    | Recommended SOC Action        | Immediate Tier 3 escalation<br>Automated traffic blocking activated<br>Formal incident declared | Critical tier response protocol triggered automatically                        |
| 7    | Output to Final Aggregator    | { component: 'UDP', score: 3120,<br>tier: 'Critical', confidence: 0.97<br>}                     | JSON packet transmitted to group Combined Scoring Ranking Table                |